Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/08/2024
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Assessing the Impact of Agr Quorum Sensing on Staphylococcus aureus Physiology in the Space Flight Environment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-0700	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	02/08/2021	**End Date:**	02/28/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	End date changed to 02/28/2025 per PI information (Ed., 1/8/24).		
Key Personnel Changes/Previous PI:	1. Dr. Wayne Nicholson (Co-I, University of Florida) was removed from this grant in summer 2023, due to his unexpected passing. 2. Dr. Robert Shields (Co-I) is now an Assistant Professor at Arkansas State University. He has been removed as University of Florida (UF) key personnel and we are in the process of setting up a subaward for him to continue his participation on this project. His expertise is required for processing samples and analyzing the TnSeq data to be generated from Aim 1 of this proposal (TnSeq on S. aureus flight and ground control-grown Tn libraries).		
COI Name (Institution):	Carroll, Ronan Ph.D. (Ohio University) Edelmann, Mariola Ph.D. (University of Florida, Gainesville) Shields, Robert Ph.D. (Arkansas State University)		
Grant/Contract No.:	80NSSC21K0601		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Effective protection of astronaut health is predicated on a thorough understanding of the effects of the space flight environment on virulence potential of bacterial pathogens. Staphylococcus aureus is a pathogen that can cause a wide variety of devastating infections, many of which result from the ability of S. aureus to grow as biofilm, its wide arsenal of secreted virulence factors, and its highly adaptive metabolism. These properties are regulated, in part, by its Accessory Gene Regulator (Agr) quorum sensing system. Analysis of S. aureus global gene expression data from a previous space flight experiment (Biological Research In Canisters (BRIC)-23; GeneLab Data System (GLDS)-145)) on the International Space Station (ISS) suggested that the Agr-mediated quorum sensing system (QSS) and expression of some of its target regulon (secreted toxins, exoproteases) were highly upregulated during space flight growth. The overall hypothesis of this proposal is that the S. aureus Agr quorum-sensing system is an important regulator of biofilm formation, physiology, and virulence factor expression when this bacterium is exposed to microgravity or other variables related to the space flight environment. To address this, S. aureus biofilm formation in a wildtype and isogenic agr-deficient mutant strain will be compared using a random positioning machine (RPM) to simulate microgravity, and an ISS space flight experiment will be conducted using 4 Biological Research in Canister Petri Dish Fixation Units (BRIC-PDFUs). One BRIC-PDFU (n=5 PDFUs + 1 HOBO temperature logger) will be used to grow a S. aureus transposon library, in order to define which gene(s) are required for thriving in the space flight environment. Two BRIC-PDFUs will be used to grow S. aureus wildtype (same strain used in BRIC-23) and its isogenic agr mutant (n=5 PDFUs + 1 HOBO temperature logger per strain), to determine the contribution of Agr quorum sensing to gene expression changes previously observed in the BRIC-23 flight samples. In a fourth BRIC, PDFUs modified to cancel the potential confounding effects of microgravity on fluid geometry, and hence aeration, in BRIC units (modifications are being developed by Co-Investigator Nicholson through a separate NASA project), will be used to grow the wild-type S. aureus strain (n=5 modified PDFUs + 1 HOBO temperature). The growth media, time, and other experimental variables will be identical to those previously used in the BRIC-23 experiment in order to directly compare results. Asynchronous ground controls for each of the four space flight BRIC-PDFUs (n=5 PDFUs + 1 HOBO temperature logger per BRIC) described above will be treated identically in the ISS Environmental Simulator at Kennedy Space Center (KSC) using the recorded flight temperature data. Upon return to Earth, flight samples and ground-based controls for the wild-type and agr mutant BRIC-PDFUs will be processed for RNA-seq, secreted proteomics, metabolite, and growth measurements. The transposon library BRIC-PDFUs will be processed for DNA isolation, Illumina sequencing, and Tn-seq analysis. These proposed experiments will allow us to make more refined predictions about the pathogenic potential of S. aureus, especially in relation to astronaut health. These experiments align with Decadal Survey Recommendation P1, sub-topics MB-B (Studies designed to determine how spaceflight conditions affect microbial biofilm formation... and their potential to cause disease) and MB-D (to use the ISS as a microbial observatory to determine foundational results... that contribute to the health of future space explorers), as well as address several guiding questions of the 2016-2025 SB Science Plan: To help identify the "underlying genetic, molecular and biochemical processes influenced by the space environment" for S. aureus (MB-1), and offer additional insight into how space flight conditions such as microgravity "...influence microbial biofilm formation and microbial community development" (MB-3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Agr is a global regulator of both virulence and metabolic genes, therefore stimulation of this quorum-sensing system in response to space flight conditions may impact the pathogenic threat of S. aureus to astronauts during long-term space flight missions, especially since space flight is known to impose a suppressive effect on both innate and adaptive immune cell responses to pathogen-related stimuli. The research proposed in this application is directed at gaining a better understanding of the contribution of Agr quorum sensing to physiological adaptation and pathogenic potential of S. aureus in response to the space flight environment. It is anticipated that the results of these proposed experiments will allow us to make more refined predictions as to which virulence properties and physiological changes are likely to occur in S. aureus during space flight, and how this may impact its pathogenic potential and effect on astronaut health.
Task Progress:	Manuscripts accepted for publication: Hauserman, MH, Ferraro, M.J., Carroll, R.K., Rice KC. 2024. Altered Quorum Sensing and Physiology of Staphylococcus aureus During Spaceflight Detected by Multi-omics Data Analysis. In press: npj microgravity (NPJMGRAV-01100R1). [January 2024 Update: Now published. See Cumulative Bibliography. (Ed., 1/12/24)] Research progress: Aim 1: ISS flight experiment to elucidate the physiological response of S. aureus to the space flight environment In year 3, we completed both Science Verification Testing (SVT) and Experiment Verification Testing (EVT) in preparation for our flight experiment (scheduled for launch on NG-20 on January 29, 2024). After SVT, it was decided to scramble the loading of the 4 experimental types/replicates amongst the four Biological Research In Canisters (BRICs), and this scrambled configuration will be used for the flight and asynchronous ground controls. All success criteria were achieved for both SVT and EVT. RNA sequencing (RNA-seq) was performed on n=4 RNA from EVT. Using cutoffs of 2-fold change in expression and unadjusted p-value < 0.05, nine genes showed upregulated expression and eighteen genes showed downregulated expression in the UAMS-1 gizmo plate cultures relative to the UAMS-1 unmodified petri plate cultures. After applying a False Discovery Rate (FDR) = 0.1 using the Benjamini-Hochberg calculation, only one upregulated gene and one downregulated gene were significant. Given that expression of very few genes was altered in this comparison, it is likely S. aureus experiences similar physiology and growth following desiccation on the unmodified petri dishes and gizmo petri dishes. Using the same cutoffs described above, six genes showed upregulated expression and twelve genes showed downregulated expression in the agr mutant petri plate cultures relative to the UAMS-1 petri plate cultures. After applying an FDR = 0.1 using the Benjamini-Hochberg calculation, no upregulated genes and nine downregulated genes were significant. These highly downregulated genes in the agr mutant are expected, as this strain lacks the genes encoding the Agr quorum sensing system, and therefore positive targets should also be downregulated. Since expression of few other genes was altered in this comparison, it is likely that both the S. aureus wildtype and agr mutant experience similar physiology and growth following desiccation on the petri dishes. Carroll subaward progress report (Ohio University): Subaward has not been used yet, as the work provided by Co-Investigator (Co-I) Carroll cannot be completed until flight and ground control samples are received. We will be

	requesting a no-cost extension on both the primary grant and this subaward. Aim 2: Determine the effect of Agr and simulated microgravity on S. aureus biofilm growth using a random positioning machine (RPM) We validated use of newly constructed superfolder green fluorescent protein (GFP) reporter plasmids (constitutive sarP1 promoter, agr P3 promoter, and promoterless GFP) in 0 g RPM, 0.9 g RPM, and 1 g (static control) biofilms in strains UAMS-1 and its agr deletion mutant (at least n=3 experiments for each strain and GFP reporter). These results validated our previously-obtained preliminary data with another agr P3 reporter construct, verifying that the agr P3 promoter is much more active when biofilms are grown under 0 g simulation compared to 0.9 g simulation and 1 g static control. These biofilms were grown in medium that physiologically represses induction of the Agr quorum sensing system. Therefore, a signal in the 0 g simulation appears to be over-riding this repression. To delve more into this mechanism, we grew planktonic cultures of UAMS-1 in biofilm media in High Aspect Ratio Vessels (HARVs) on the RPM (0 g) and in static (1 g) control HARVs (n=4 each). We are currently isolating RNA from these cultures which will be subjected to RNA-seq analysis.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Articles in Peer-reviewed Journals	Hauserman, M.R., Ferraro, M.J., Carroll, R.K. and Rice, K.C. "Altered quorum sensing and physiology of Staphylococcus aureus during spaceflight detected by multi-omics data analysis." npj Microgravity. 2024 Jan 8;10(1):2. https://doi.org/10.1038/s41526-023-00343-7 ; PMID: 38191486; PMCID: PMC10774393 , Jan-2024